Agenda

* Overfit / underfit code
* Bias / variance Tradeoff
* ** Regularisation — L1
  L2
  $L_1 + L_2$

Made with Goodnotes

Recap

* Variants of GD
* polynomial regression
* underfit & overfit → 3 { M_1, M_2, M_3

Made with Goodnotes

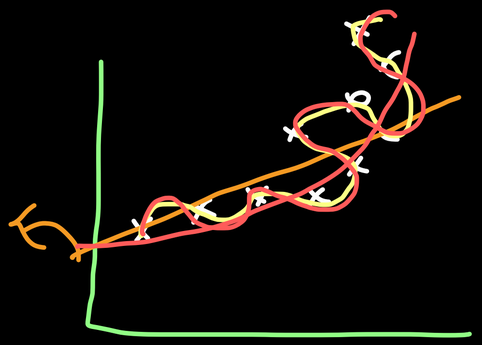
← performance →

	Train	Test (unseen)
underfit	poor (low)	poor (low)
perfect fit	Great	Great
overfit	very good (Best)	Moderate (avg)

Simple model

↓

Complex model



f_1^2 f_1^3 f_1^4

Understanding Generalisation :

Comparison of these three models M1, M2, M3 gave very important concept of :

Generalization-

If there are several ML model, Generalization states that:

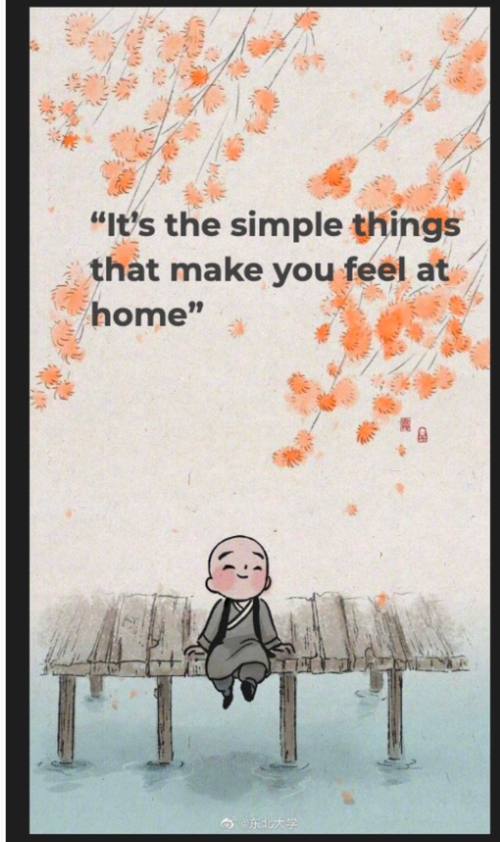
- Always choose the models that learns and understand the data inside out so that it can make good predictions on unseen data.

understand pattern



i.e test performance high

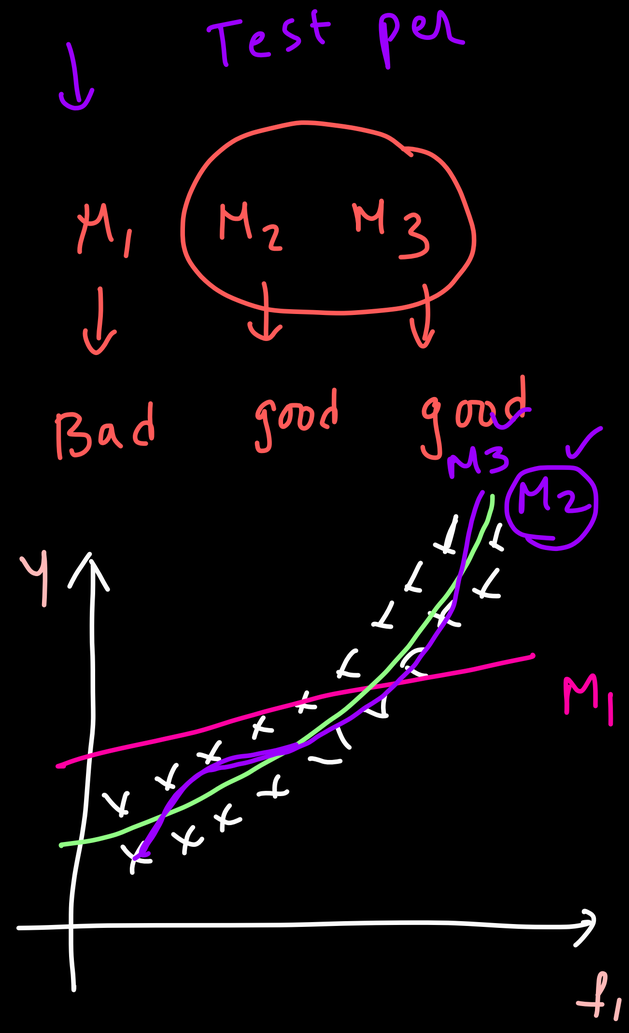
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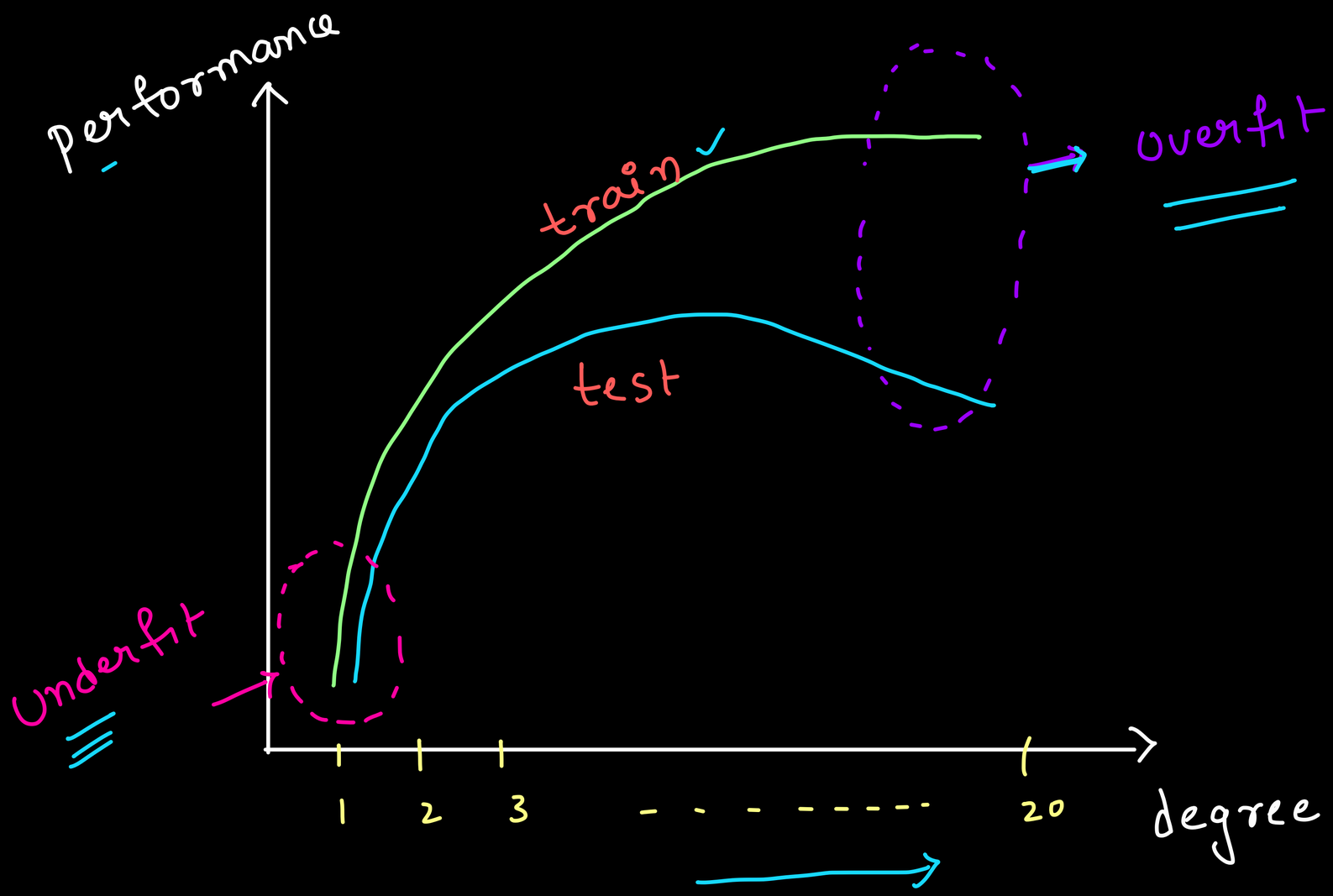


OCCAM's RAZOR

There is another rule that we follow while choosing best model.

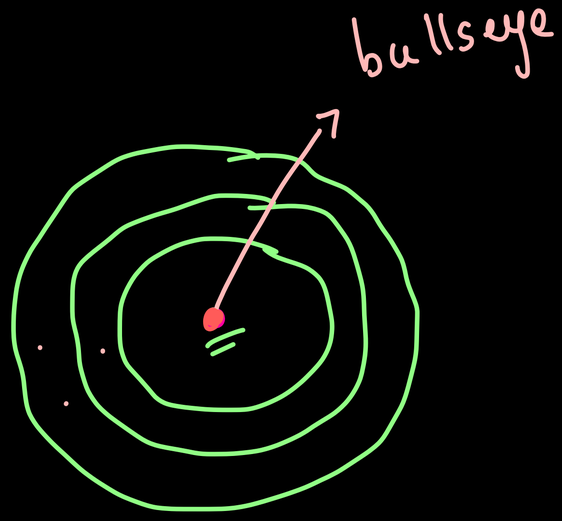
> "There are many solutions to the problem,
> always choose the simpler one"




Performance
train
test
Overfit
Underfit
1
2
3
20
degree

Bias | variance Trade off

Archery



Since she started learning she is very chaotic and just hitting arrow anywhere randomly.



This is the case of,

HIGH BIAS	High loss, since target is missed everytime.
HIGH VARIANCE	Because she is not consistent, just hitting ramdomly

vary

Betty tried hard to learn, finally she can now hit near the bull's eye. But she lacks consistency

CASE II



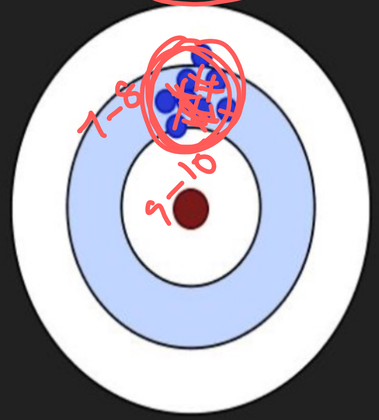
This is the case of,

✓ LOW BIAS	Because she is hitting near target.
HIGH VARIANCE	Due to lack of consistency.

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So she decided to work on consistency with time, she learnt the art of consistency but was still missing target.

CASE III



This is the case of,

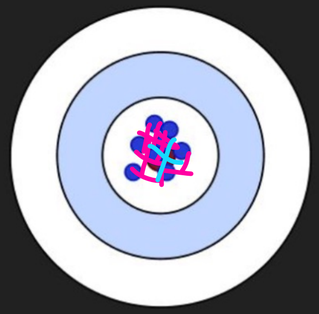
HIGH BIAS	High loss, since target is missing
LOW VARIANCE	because shots are consistence

→ high bias
low variance

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Finally, after years of practice she mastered the art of archery.

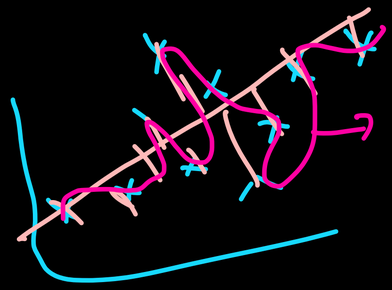
CASE IV



This is the case of,

LOW BIAS	To the bull's eye
LOW VARIANCE	Consistent

→ { low bias
low variance



low bias

✓ Bias :

error $(y - \hat{y})$

↳ high bias

captured the pattern

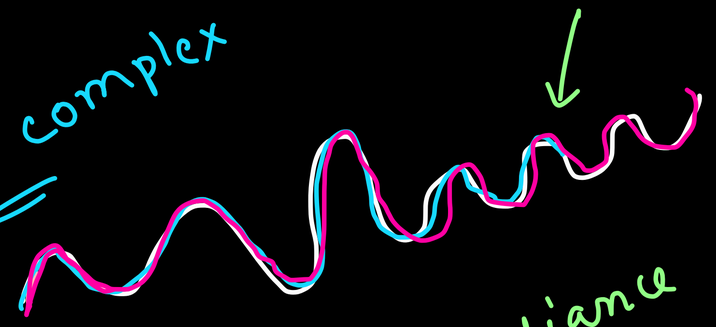
low bias

✓ Variance ✓

low variance ✓

high

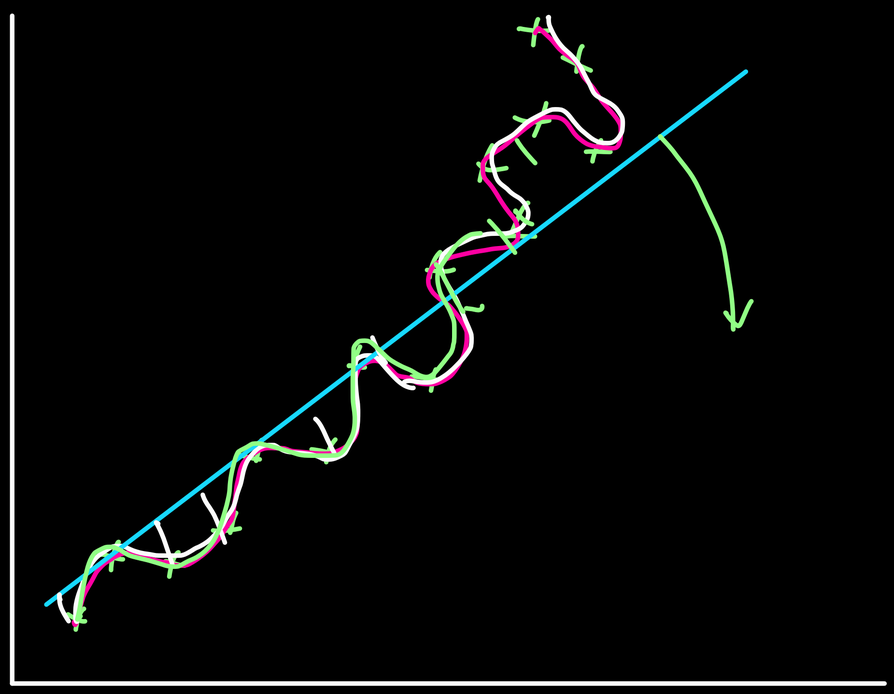
too complex



high variance

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overfit → variance → high
Bias → low



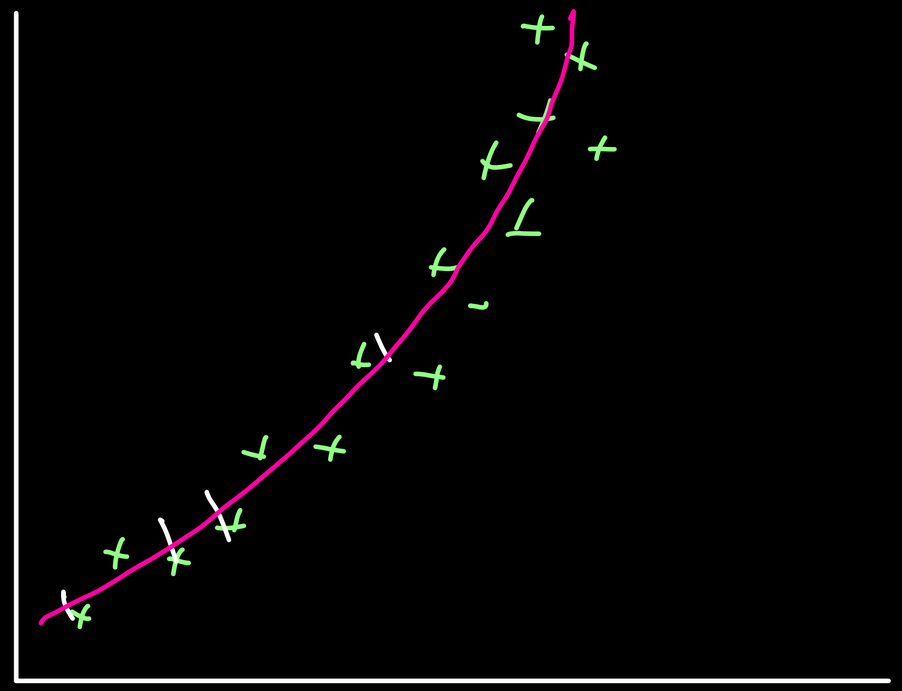
→ overfit
variance ✓
Bias ↓

underfit :

variance → low
Bias → high

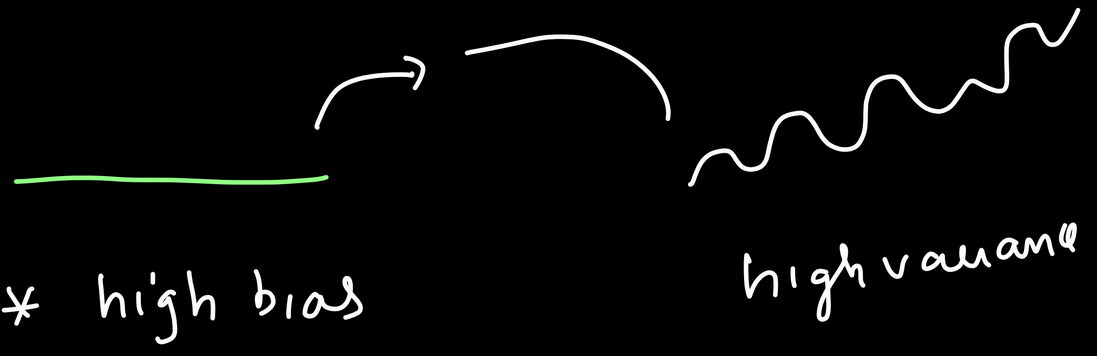
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perfect fit model



→ { low bias, low variance } ✓

→



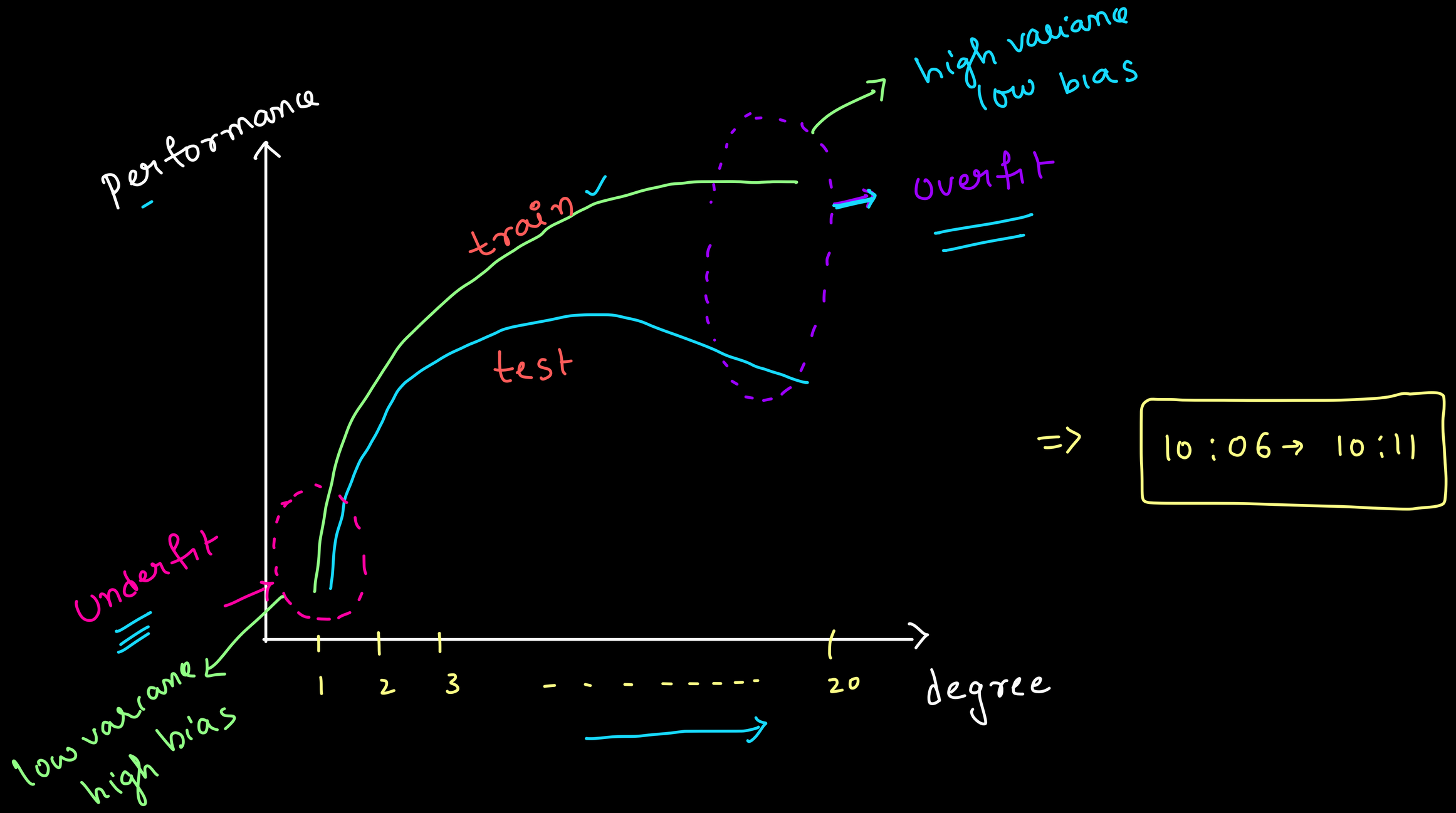
* high bias

high variance

→ low bias

low variance

Made with Goodnotes

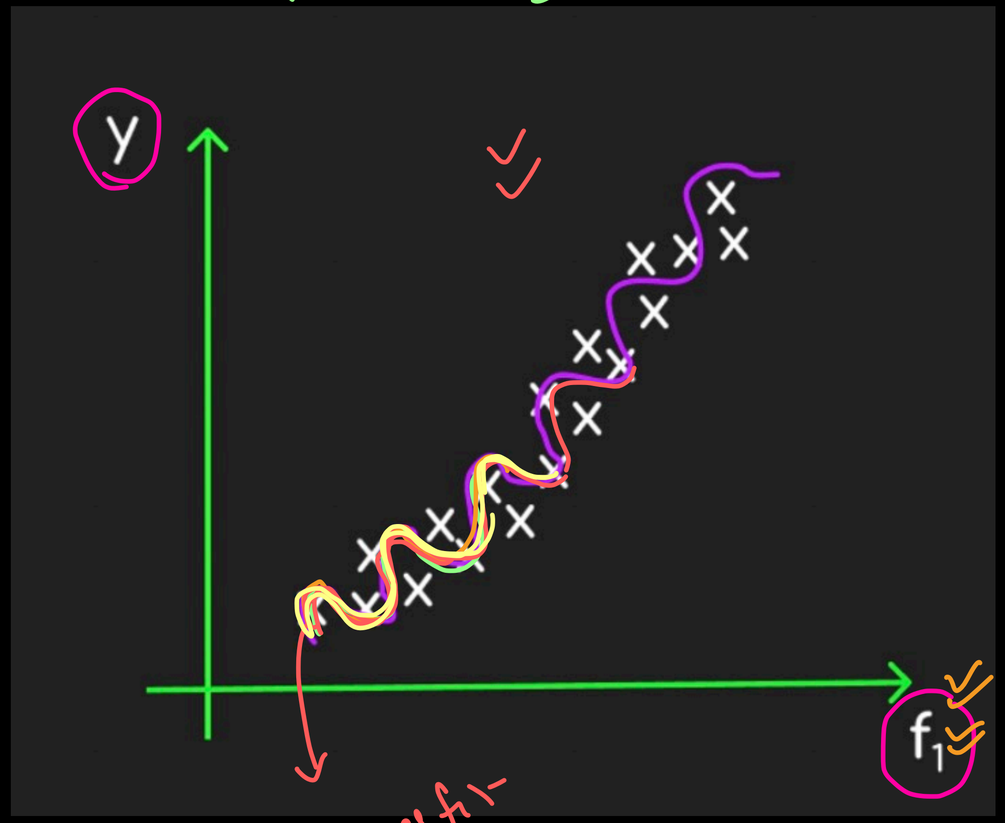

Performance
train
test
high variance
low bias
overfit
Underfit
low variance
high bias
1
2
3
20
degree

=> 10:06 → 10:11

Regularization

→ Solution to overfit / underfit.

training data



overfit-

complexity → high

Bias → low

variance → high

underfit / overfit → overfit

Made with Goodnotes

$$\hat{y} = w_0 + w_1 f_1 + w_2 f_1^2 + w_3 f_1^3 + w_4 f_1^4 + w_5 f_1^5$$

Reduce influence of these higher order features

$$w_3 = w_4 = w_5 = 0$$

$$\hat{y} = w_0 + w_1 f_1 + w_2 f_1^2 \rightarrow \text{low bias, low vari}$$

New Loss function = MSE ↑↑↑

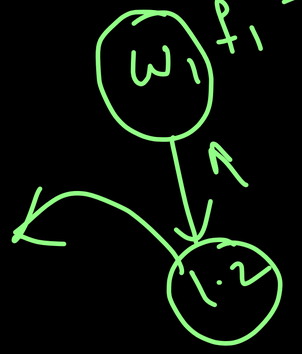
$$\frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2 + \sum_{i=1}^{d} w_0^2$$

MSE

$\rightarrow w_1^2 + w_2^2 + \cdots + w_5^2$

regularisation parameter

$w_1 \quad w_2 \quad w_3 \quad \cdots \quad 0$

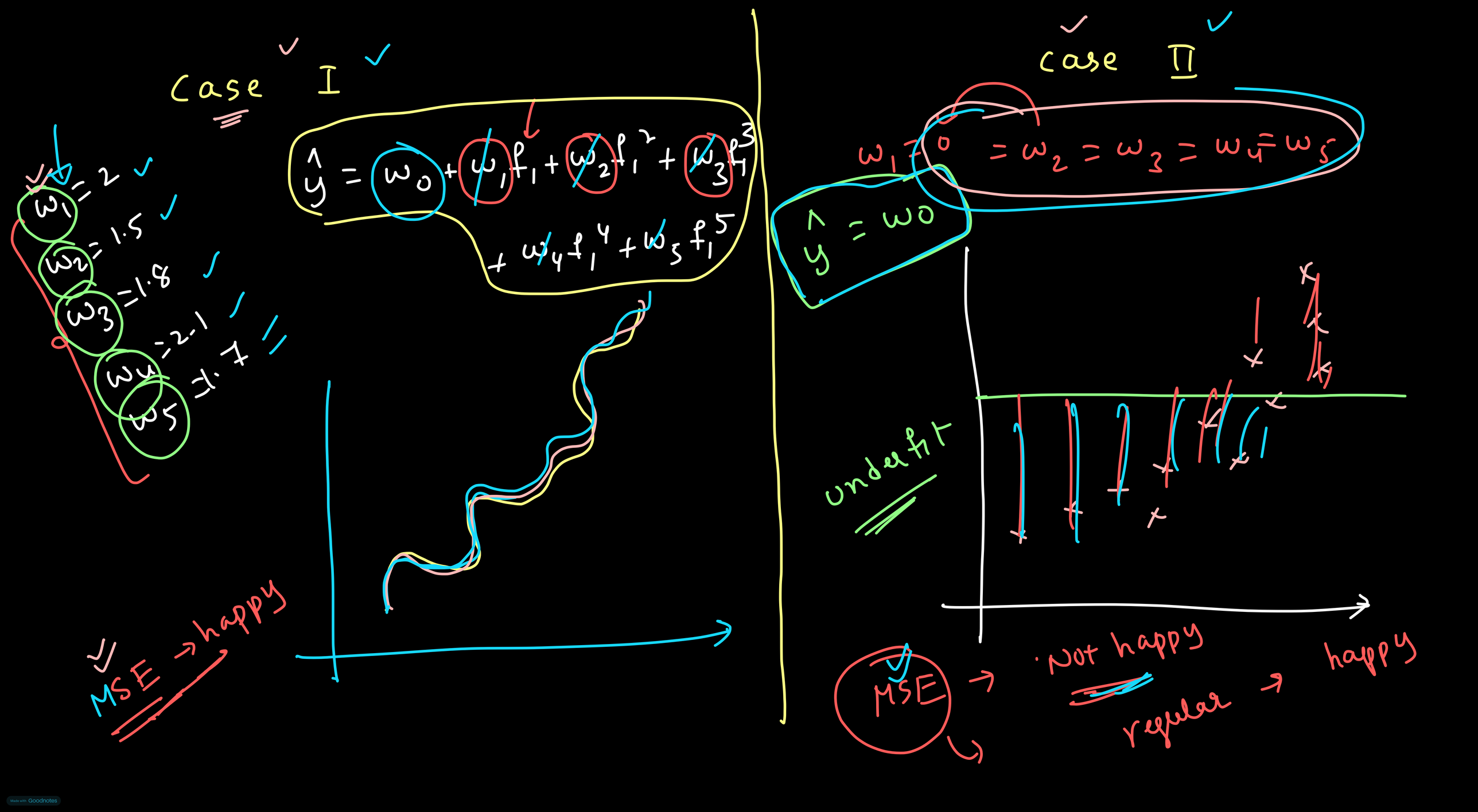


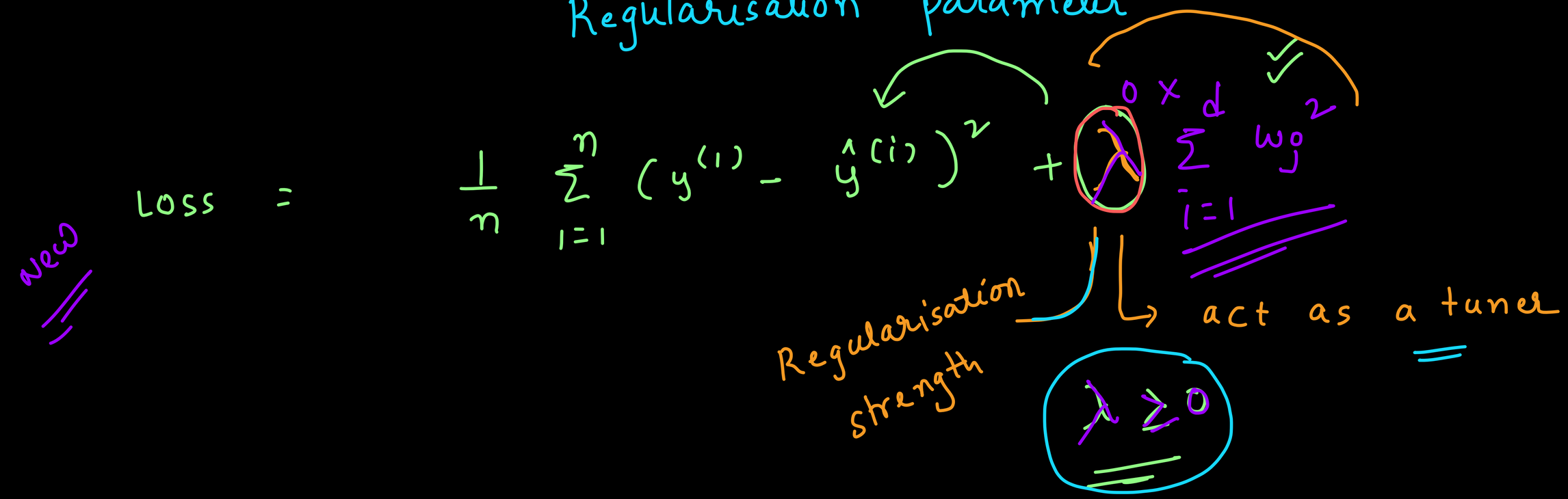
$f_1 + w_2 + w_3$

2.1 0.8

$1.2^2 \quad 2.1^2 \quad 0.8^2$))

$\hat{y} = w_0$


Case I
$\hat{y} = w_0 + w_1 f_1 + w_2 f_1^2 + w_3 f_1^3 + w_4 f_1^4 + w_5 f_1^5$
$w_1 = 2$
$w_2 = 1.5$
$w_3 = 1.8$
$w_4 = 2.1$
$w_5 = 1.7$
MSE → happy
Case II
$w_1 = 0 = w_2 = w_3 = w_4 = w_5$
$\hat{y} = w_0$
underfit
MSE → not happy
regular → happy


Regularisation parameter
new
Loss =
$\frac{1}{n} \sum_{i=1}^{n} (y^{(i)} - \hat{y}^{(i)})^2 + \lambda \sum_{j=1}^{d} w_j^2$
Regularisation strength
act as a tuner
$\lambda \geq 0$

$$\hat{y} = w_0 + w_1 f_1 + w_2 f_1^2 + w_3 f_1^3 + w_4 f_1^4 + w_5 f_1^5$$

$\lambda = 0$ small

$\Rightarrow$ → there is no importance to reg

→ no penality on weights

MSE ↓

overfit

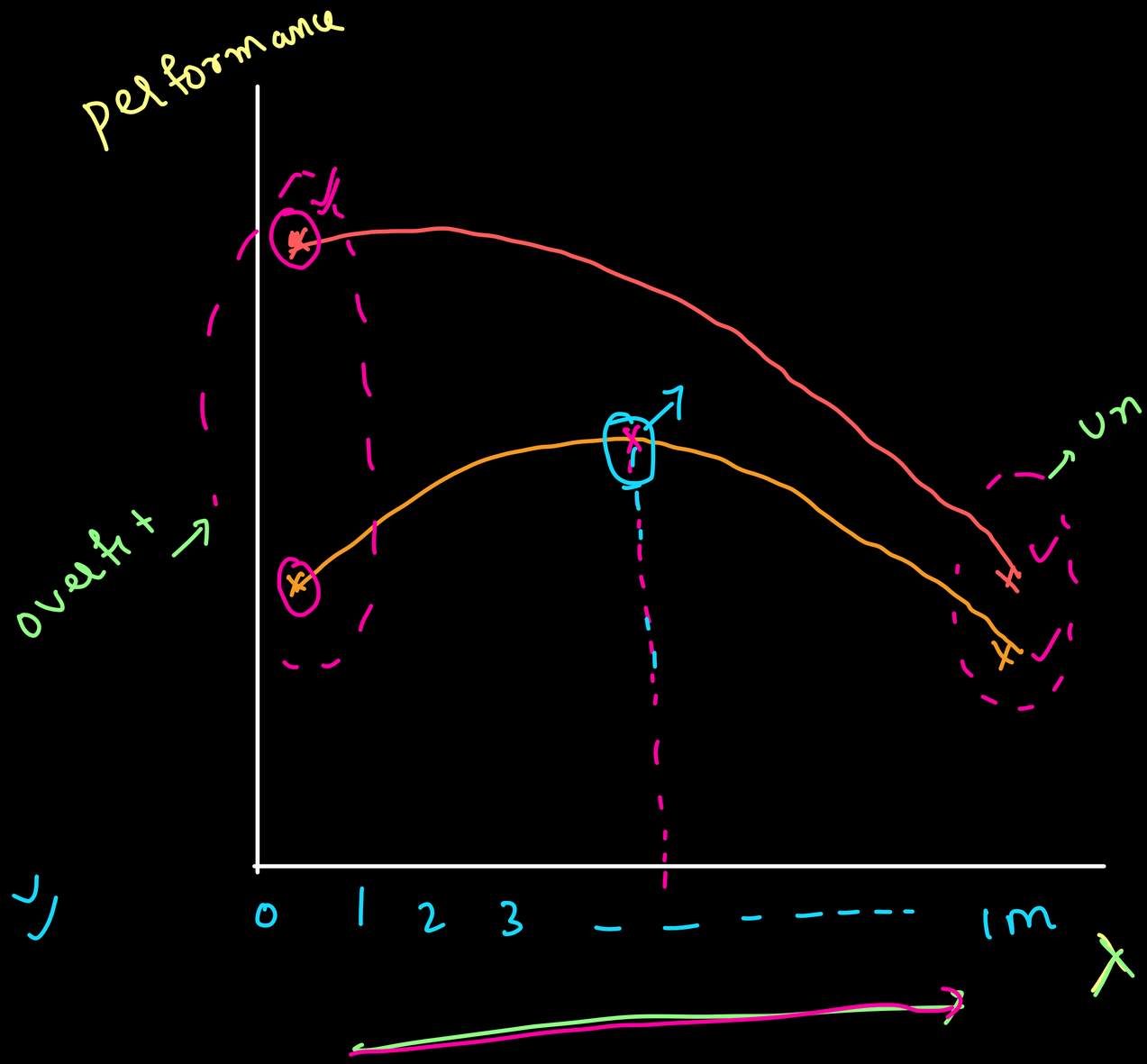
$\lambda = 10^8$ v large

→ more importance to reg

$\hat{y} = w_0$

$w_4, w_5 \to 0$

underfit



train test

$\lambda = 0$ (high, low)

$\lambda = 10^8$ (low, low)

→ How it decides which features | w's to make 0?



new loss = MSE + λ · Regularisation

$w_0 + w_1 f_1 \rightarrow$

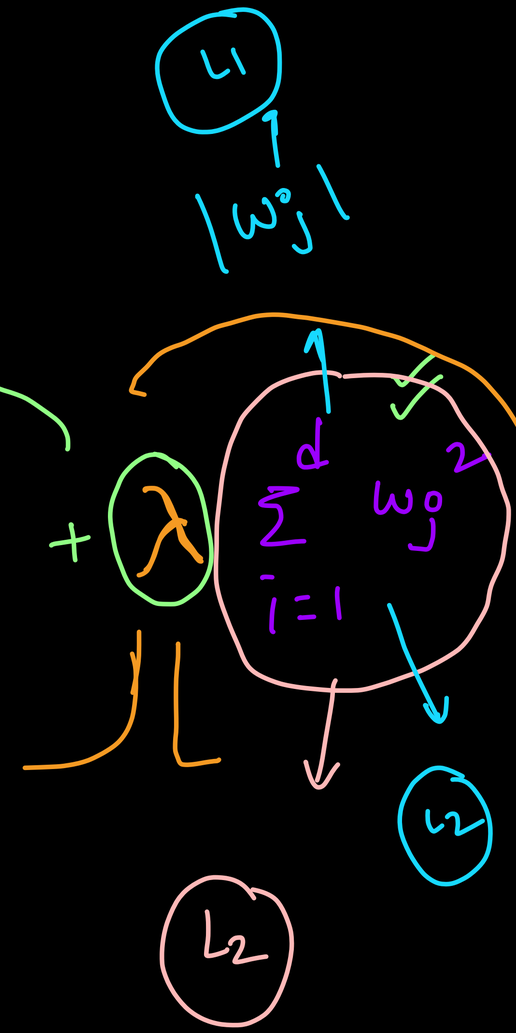
2 types

1. L_1 Regu
2. L_2 Regularisation

Norm of a vector →

$$w = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

Loss $= \frac{1}{n} \sum_{i=1}^{n} (y^{(i)} - \hat{y}^{(i)})^2 + \lambda \sum_{j=1}^{d} w_j^2$



eclidian

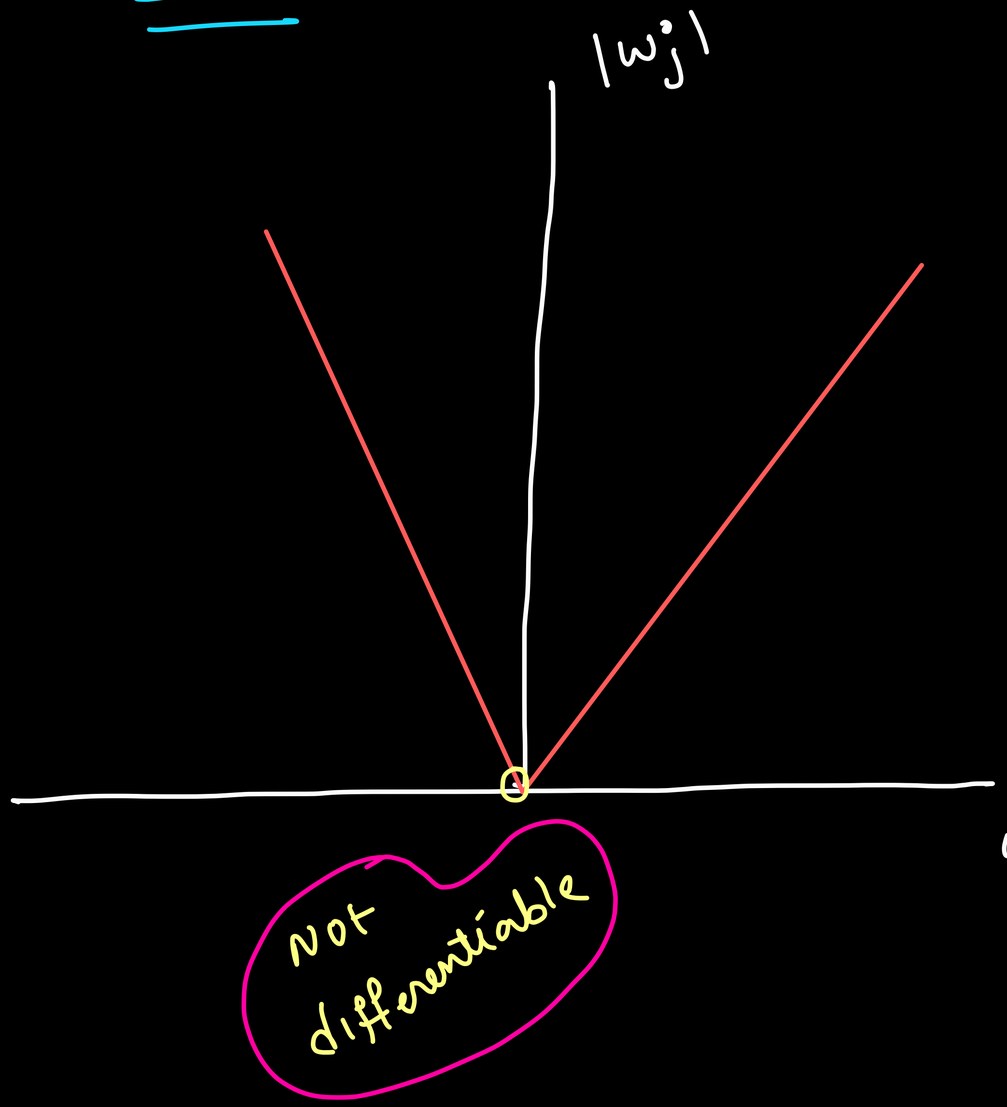
$$\|w\|_2 = \sqrt{w_1^2 + w_2^2 + w_3^2}$$

manhaullen

$$\|w\|_1 = |w_1| + |w_2| + |w_3|$$

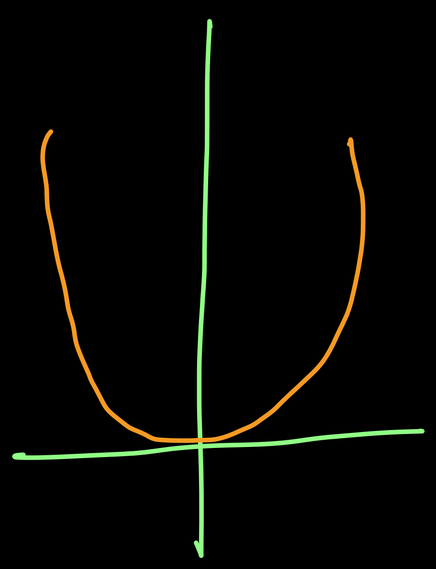
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✓✓ L_1 Regularization $|w_j|$



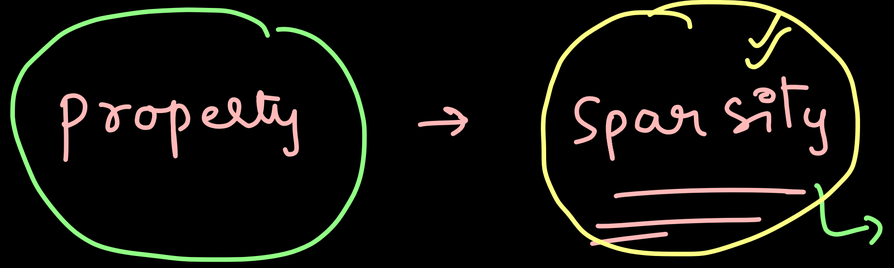
$$\frac{\partial |w_j|}{\partial w_j} = \begin{cases} +1 & w_j > 0 \\ -1 & w_j < 0 \\ \simeq 0 & \\ w_j = 0 & \end{cases}$$

hack



$$\frac{\partial w_j^2}{\partial w_j} = 2w_j$$



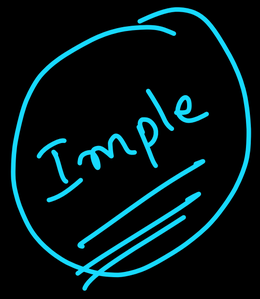


lot of weights become 0

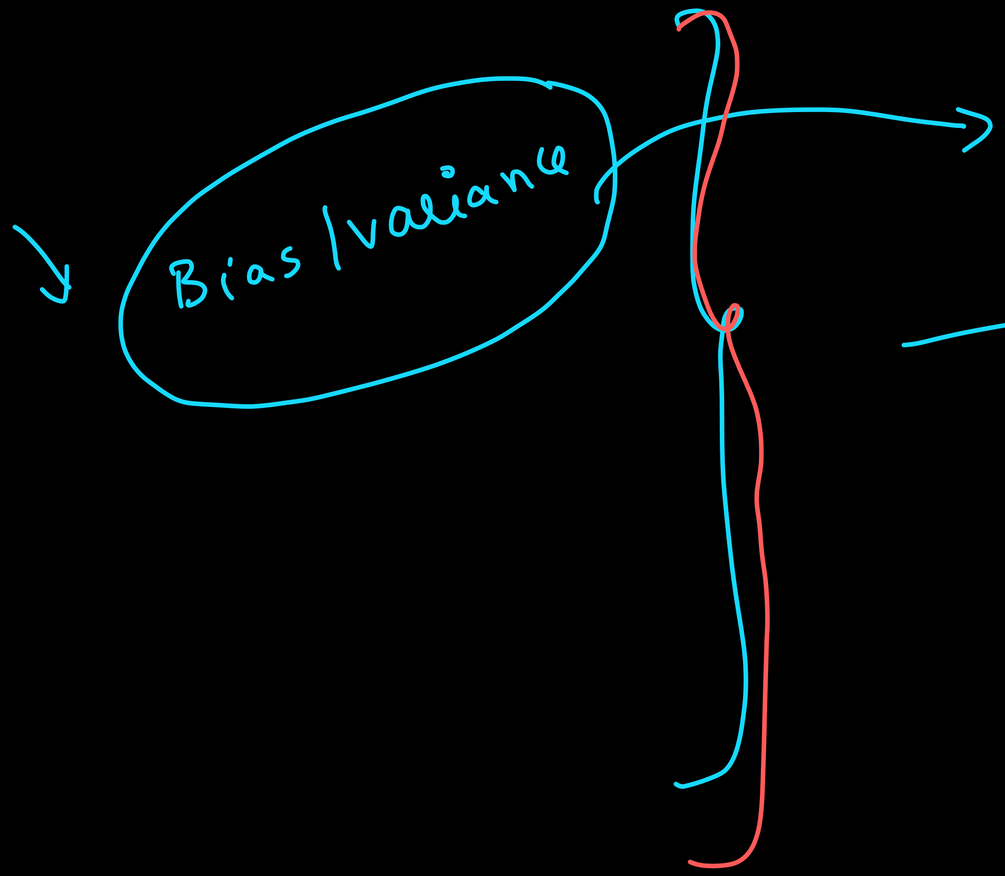


→ no sparsity

↳ weights will be closer to 0 but not ≠0



⇒ L_1+L_2 → Cross validation



$$\text{Loss} = \frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - \hat{y}^{(i)}\right)^2 + \lambda_1 \sum_{j=1}^{d} |w_j| + \lambda_2 \sum_{j=1}^{d} w_j^2$$

Lasso

Ridge

Made with Goodnotes